Model Theory

Overview

These pages describe how SWAT+ represents the processes that move water, sediment, nutrients and other constituents through a watershed. Each page covers one group of processes: the equations the model solves, the input files and columns that control them, and where SWAT+ 62.0.1 differs from the SWAT 2009 theoretical documentation.

The landscape processes run once per day for every hydrologic response unit (HRU): weather, snow, evapotranspiration, surface runoff, soil water, erosion, plant growth and the soil nutrient cycles. Their results leave the HRU as surface runoff, lateral flow, tile flow and percolation, and pass to the routing objects: channels, aquifers, reservoirs and wetlands, and recall points. Decision tables apply management and reservoir release rules across both groups.

Pages

Group	Page	Contents
Landscape water	Overview and water balance	HRU water balance, the order of daily calculations, where each term is reported
Landscape water	Climate and weather	Measured and generated weather, day length, sub-daily precipitation, elevation lapse, potential evapotranspiration
Landscape water	Hydrology	Canopy interception, surface runoff, lateral flow, percolation, tile flow, lag to the channel
Landscape water	Soil water and temperature	Soil layers, water retention, initial soil water, soil evaporation, soil temperature
Landscape water	Erosion and sediment	Peak runoff rate, MUSLE, sub-daily erosion, sediment in lateral flow, urban loads, filter strips, grassed waterways and user BMPs
Routing	Channels	Channel routing, sediment transport and in-stream water quality
Routing	Aquifers	Aquifer objects and the gwflow groundwater module
Routing	Reservoirs and wetlands	Reservoir water balance and release rules, wetlands
Routing	Recall and point sources	Recall time series and point source loads
Routing	Decision tables	Condition evaluation and actions
Plants and chemistry	Plant growth and land use management	Heat units, biomass, leaf area, stresses, harvest, scheduled management
Plants and chemistry	Nutrient cycling	Soil nitrogen and phosphorus pools, transformations, losses; algae, CBOD and oxygen in runoff; septic systems
Plants and chemistry	Carbon	Static and CENTURY-based soil carbon
Plants and chemistry	Pesticides	Application, partitioning, decay and transport
Plants and chemistry	Salts and constituents	Salt ion chemistry, other constituents, pathogens and heavy metals

SWAT 2009 processes not in SWAT+ 62.0.1

The SWAT 2009 theoretical documentation describes some processes that SWAT+ 62.0.1 does not simulate, or simulates in a different way:

- Elevation bands within a subbasin. SWAT+ corrects precipitation and temperature for the elevation of each object instead (see Climate and weather).

- Monthly climate change adjustments of precipitation, temperature, solar radiation and humidity. SWAT+ 62.0.1 has no input for them; adjust the weather files instead.
- Hourly air temperature and the distribution of solar radiation over the hours of the day.
- The adjustment of the curve number for slope.
- Transmission losses from tributary channels within a subbasin. SWAT+ has no tributary channels; losses are computed for each channel object.
- Bank storage in the main channel.
- The SWAT 2009 channel sediment routing (simplified Bagnold stream power equation, channel erodibility and cover factors, downcutting and widening). SWAT+ uses the channel sediment model described in Channels.
- Potholes and other depressional areas, replaced by wetlands and paddies in HRUs (see Reservoirs and wetlands).
- Reservoir release from measured monthly outflow or a target storage. SWAT+ uses release decision tables and measured releases from recall files.
- Chlorophyll-a and Secchi depth relations for lakes and reservoirs.
- The stream and dam reaeration formulas of QUAL2E; SWAT+ uses the reaeration rate given in `nutrients.cha`.
- The two-population bacteria model (persistent and less persistent bacteria), replaced by the pathogen routines described in Salts and constituents.
- Continuous fertilization and automatic fertilization by nitrogen stress. Decision tables can fertilize in response to conditions instead (see Decision tables).
- Consumptive water use and water transfer between objects, replaced by water allocation (see Water allocation).

Page layout

Every symbol is defined under its equation. Where a symbol is a user input, the input file and the column header the SWAT+ Editor writes are named with it, and the basin switches and parameters that change the result are listed together on each page.

The last section of every page, Sources, is for developers. It links to the Fortran routines at the 62.0.1 tag, maps symbols and columns to Fortran variable names, and records implementation details.

Sources

- Daily HRU sequence: `hru_control.f90`
- Object loop that calls HRUs, channels, aquifers and reservoirs: `command.f90`

The pages in this section were checked against SWAT+ 62.0.1 (commit `6f51536`). Where a page and the Fortran source disagree, the source describes what the model does.